

Arup Anil Chakravorty

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SUMMARY

Full-stack engineer with **6+ years** shipping production systems for **millions of users** across OTT, social, e-commerce, and GIS marketplace products, owning delivery end to end from AWS infrastructure through React, Flutter, and Java/Python microservices. Currently **founding engineer** building a multi-service consumer platform from zero, with depth in **LLM and agentic systems**: LangGraph orchestration, hybrid retrieval, guardrails, and human-in-the-loop approval design.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, Dart, SQL

Frontend: React, Next.js, Flutter (Android + iOS), Vite, HTML/CSS

Backend: Spring Boot, FastAPI, Node.js, REST APIs, Microservices, JWT Auth

AI / ML: LangGraph, LangChain, OpenAI/OpenRouter APIs, RAG, Hybrid Retrieval (BM25 + RRF), LLM Guardrails, Offline Eval Harnesses, Image Detection

Cloud, Data & DevOps: AWS (EC2, S3, Lambda, RDS, CloudFront), Docker & Compose, CI/CD, Git, Nginx, Firebase, PostgreSQL, MySQL, SQLAlchemy, Alembic, Query Optimization, pytest

WORK EXPERIENCE

Guchhait Software Solutions Pvt. Ltd.

Jul 2026 – Present

Founding Engineer — TasteDash (cross-platform food delivery)

- Joined as **founding engineer** with no codebase or infrastructure in place; defined the Java Spring Boot and Python **microservice decomposition** behind clean REST contracts, plus the containerized environments and build checks the team now works on, so ordering, catalog, and delivery ship independently rather than as one blocking monolith.
- Owned the full mobile surface for a team too small to staff two native clients; shipped a **single Flutter codebase to both Android and iOS**, holding every feature at one implementation rather than two.

Sunyam Software LLP

Aug 2019 – Nov 2025

Senior Full-Stack Engineer

- Re-architected the AWS footprint of a global OTT and social platform that had outgrown its design, adding autoscaling, edge caching, and SQL tuning to **cut latency 40%** at **99.9% uptime** for millions of users.
- Manual, error-prone deploys capped release velocity; built **Git-based CI/CD pipelines** with automated testing and staged rollouts, **shortening release cycles 60%** and making zero-downtime deployment the default.
- With no in-house ML capability on the team, **trained image-detection models** and built **agentic LangGraph + OpenAI applications** that automated classification and decision workflows previously done by hand.
- Mentored **5+ junior engineers** on full-stack development, system design, and cloud architecture while acting as technical point of contact for requirements and demos.

PROJECTS

TrustDesk — AI-First Customer Support Operations Platform

🌐 trustdesk.dexcode.app

Java 21, Spring Boot 3, FastAPI, LangGraph, React 19, MySQL, Docker Compose

- Architected **three independently deployable services** (Spring Boot system-of-record, FastAPI AI/retrieval service, React 19 frontend) behind single-origin Nginx routing, with no backend, database, or AI port exposed to the host.
- Designed the AI as **recommend-only**: the backend alone executes sensitive tool actions, and only after human approval, with retries absorbed by a **database-level idempotency constraint**, so a duplicate submit returns the existing action instead of executing it twice.
- Built **hybrid retrieval** fusing BM25 with embedding similarity via Reciprocal Rank Fusion, degrading to BM25-only on provider failure rather than downing the service; a labeled eval harness over the *real* pipelines scored **100% unsafe-action block rate, citation coverage, and triage accuracy** across 8 cases including **3 prompt-injection attacks**.

DermaCare AI — Skin Analysis & Personalized Report Backend

FastAPI, LangChain, OpenRouter, SQLAlchemy 2.0, PostgreSQL, Firebase

- Built the FastAPI backend for a skin-scan mobile app: Firebase ID-token auth, Storage-backed uploads, and rate limiting fronting a **two-stage LangChain pipeline** over OpenRouter — a vision model detects conditions, a text model writes the report — each stage **swappable by config**.
- Modeled data in SQLAlchemy 2.0 with Alembic migrations on PostgreSQL, moved scans **off the request path** so uploads never block on two LLM round-trips, and made the suite runnable with **zero credentials or network calls** by stubbing Firebase and both LLM stages over SQLite.

EDUCATION

Backend Engineering Launchpad, Airtribe (2025–26) | **B.Com**, Calcutta University (2016–19)